Journal of Experimental Criminology
https://doi.org/10.1007/s11292-026-09731-x

AI-generated human stimuli for experimental social science

Chandler G. Robinson[1] · Ian T. Adams[2] · Matthew W. Logan[3] · J. Pete Blair[3]

Received: 30 October 2025 / Accepted: 6 January 2026
© The Author(s), under exclusive licence to Springer Nature B.V. 2026

Abstract

Objectives: This study evaluates whether AI-generated police officer profiles can serve as valid, reproducible visual stimuli for experimental criminology. We test whether systematically varied images differing in race, gender, and body composition convey intended traits and elicit theoretically coherent impressions of competence, agency, and warmth.

Methods: Using a $4 \times 2 \times 3$ factorial design, we generated 24 AI-based officer profile cells (72 individual profiles) and collected 2,565 evaluations from a nationally representative U.S. sample (n = 513). Participants identified each officer's race, gender, and body type, detected one-at-a-time manipulations, and provided impression ratings across key social perception dimensions.

Results: Respondents accurately recognized all intended attributes and reliably detected manipulations across otherwise identical images. Impression ratings reproduced established patterns, including gender-linked warmth and extended body-type effects on competence and agency. Muscular police officers experience perceptual benefits while heavy officers experience a range of penalties.

Conclusions: AI-generated officer images provide a transparent, scalable, and valid framework for visual experiments in policing, criminology, and broader social science.

Keywords AI-generated images · experimental criminology · police perception · somatotype · factorial design · social judgment

✉ Chandler G. Robinson
chandler.robinson@utah.edu

Ian T. Adams
ian.adams@sc.edu

Matthew W. Logan
mwl39@txstate.edu

J. Pete Blair
pete.blair@txstate.edu

[1] University of Utah, 201 Presidents Circle, Salt Lake City, UT 84112, USA

[2] University of South Carolina, 1507 Greene St, Columbia, SC 29208, USA

[3] Texas State University, 601 University Dr, San Marcos, TX 78666, USA

Published online: 19 February 2026

Introduction

Visual experiments across the social and behavioral sciences depend on images that are consistent, transparent, and reproducible. Yet most available photo sets are scarce, idiosyncratic, and difficult to share, which constrains design space and limits cumulative inference.

Recent reviews articulate both the promise and the peril of AI in social science. Grossmann et al. (2023) argues that large language models can simulate human judgments at scale, which accelerates hypothesis testing, but also heightens concerns about transparency and bias management. Bail (2024) likewise contends that generative AI can furnish realistic, scalable stimuli for experiments and surveys, yet cautions about inherited data biases, replication challenges, and unresolved ethical issues. We respond to this agenda by validating AI-generated images of police officers as experimental stimuli. Here we show that synthetic profiles varying race, gender, and body composition pass perceptual validity checks under stimulus sampling, and they elicit theory-consistent patterns on competence, agency, and warmth across 2,565 profile ratings in a U.S. sample. The result is a transparent, reproducible workflow for large, systematically varied image sets.

Across politics, the military, policing, and organizational hierarchies, thin-slice judgments from faces systematically shape who rises to leadership. In elections, rapid impressions of competence from candidate portraits predict outcomes above chance in the United States and across Europe (Todorov et al., 2005; Antonakis & Dalgas, 2009; Berggren et al., 2010; Laustsen, 2014; Olivola & Todorov, 2010a, b), and attractiveness[1] or appearance explains vote shares even net of party and incumbency in proportional systems (Berggren et al., 2017). In hierarchical organizations, classic evidence from West Point graduation portraits shows that facial dominance forecasts rank decades later, controlling for early performance (Mueller & Mazur, 1996). Adams et al. (2024) demonstrate that officers' embodiment has concrete institutional consequences. Using archival academy headshots, they show that public judgments of facial traits and perceived leadership predict above chance which cadets later become sergeants and lieutenants. Because those leaders set policy, hiring, discipline, and organizational culture, the study links individual-level impression formation to the reproduction of institutional leadership in one of society's most consequential organizations. Here, these findings establish that embodied visual cues exert replicable effects on leadership selection across institutions.

Policing provides a critical extension of this logic and a stringent test case for adding what prior studies often omit: body composition as a core cue alongside race and gender. Research on weight stigma and embodiment shows that body size shapes judgments of competence, professionalism, and authority, and that these effects may be deeply gendered and racialized (Roehling, 1999; Puhl & Brownell, 2006; Strings, 2019; Bordo, 1993). Thus, body composition operates not merely as attractiveness, but as a status cue that intersects with race and gender in shaping social perception.

At the same time, experimental work that manipulates officer profiles yields mixed results. Some studies find that Black officers are evaluated more favorably than White officers under similar conditions (Levin & Thomas, 1997; Pica et al., 2020), others report the opposite (Castaneda, 2018; Ewanation & Maeder, 2023), and some find no main effects of officer race (Salerno & Sanchez, 2020). Prior work on stereotype content suggests that per-

[1] We recommend seeing Yonce (2014); Gordon et al. (2013) for more on "attractiveness privilege"

ceptions of warmth reflect inferred intent (whether a target is seen as cooperative or threatening) whereas competence reflects perceived capacity to act (Fiske et al., 2002; Fiske, 2018). In institutional contexts such as policing, White officers may be more readily perceived as representatives of state authority and social control, which can dampen warmth evaluations while leaving competence judgments unchanged. By contrast, officers of color may benefit from counter-stereotypical expectations of empathy or community orientation, producing higher warmth ratings without corresponding differences in competence or agency. More nuanced designs show conditionality: perceptions depend on intersections with professional conduct records (Dunbar et al., 2024) and, in some contexts, the blue identity of policing may overshadow race (Weitzer, 2000). Heterogeneity may reflect contextual moderators and methodological limits, as many studies rely on vignettes or small, idiosyncratic photo sets that constrain consistency, generalizability, and reproducibility.

We build on the leadership-perception literature by addressing the identification and reproducibility problems directly. A critical omission in this comparative literature is body composition. Social and organizational research shows that variation in somatotype systematically biases evaluations of competence, agency, and desirability (Levine & Schweitzer, 2015; Pingitore et al., 1994; Roehling, 1999; Puhl & Brownell, 2006), while formidability cues are central to status allocation and dominance in groups (Blaker & van Vugt, 2014; Lukaszewski et al., 2016; Zeng et al., 2022). We therefore treat body composition as a first-order visual attribute on par with race and gender. We develop and validate AI-generated officer profiles that (i) can be systematically varied to isolate race, gender, and body composition; (ii) pass perceptual validity checks under stimulus sampling; and (iii) elicit impression patterns consistent with established theories of social perception. This framework moves the field from observational forecasting with archival images to controlled experimentation with transparent, scalable visual stimuli, enabling cumulative social science and clarifying how embodied cues shape public perceptions in policing and, by extension, other sectors of public life.

Characterizing officer profiles in visual experiments

Visual stimuli enable experiments to manipulate social cues with high ecological face validity. In policing applications, images can capture subtle, context-specific signals that shape perceptions of professionalism, competence, and accountability. The challenge is measurement: stimuli must convey intended attributes without introducing uncontrolled confounds, and they must be reproducible across studies.

Recent advances in AI image generation (Bail, 2024; Bird & Lotfi, 2024) make it feasible to produce realistic (Cooke et al., 2025; Park et al., 2024; Epstein et al., 2023; Velásquez-Salamanca et al., 2025; Bail, 2024), systematically varied human profiles under tight experimental control (Zamudio et al., 2025). Unlike stock or staged photos, synthetic profiles allow researchers to fix nuisance features while manipulating focal traits (e.g., race, gender, body composition). This substantially expands the design space and improves consistency across treatment conditions. Because construct validity cannot be assumed, we explicitly validate that participants (i) recognize intended traits and (ii) detect targeted changes across matched images. In this

study we manipulate race, gender, and *body composition*–operationalized as somatotype (ectomorph, mesomorph, endomorph)–as core visual attributes.

Why include body composition? Much work on social evaluation focuses on race and gender, but evidence from social psychology shows that body composition systematically shapes judgments of competence, agency, and warmth. Studies document penalties associated with weight and related morphology in evaluative settings (Levine & Schweitzer, 2015; Pingitore et al., 1994; Roehling, 1999; Puhl & Brownell, 2013, 2006), alongside status-linked stereotypes tied to gender and race (Mize, 2025; Heiserman, 2023; Mize & Manago, 2018; Ridgeway, 2019; Melamed et al., 2019; Jimeno-Ingrum et al., 2009). For example, men are often perceived as more competent than women across contexts (Mize, 2025; Heiserman, 2023; Ridgeway, 2019; Ebert et al., 2014), and perceived competence varies across racial/ethnic categories (Heiserman, 2023). Treating body composition as a first-order cue (rather than a nuisance) aligns the design with this broader evidence.

Physical formidability as a status cue. Contemporary social psychology links perceived physical formidability to status allocation and dominance. We use somatotype strictly as a visible morphology for perception[2] (Blaker & van Vugt, 2014; Lukaszewski et al., 2016; von Rueden, 2014), and documents strong implicit associations between perceived size and rank (Holbrook & Fessler, 2015; Duguid & Goncalo, 2011; Hwang & Matsumoto, 2014; Yap et al., 2013; Zanolie et al., 2012). These mechanisms are germane to policing, where physical presence and fitness are salient in public expectations and task demands (Marins et al., 2018). Notably, objective fitness levels of officers do not consistently exceed those of the general public across core domains (Marins et al., 2019), which heightens the relevance of *perceived* formidability for social judgments.

Organizational norms and perceptions. Although police leaders overwhelmingly endorse the importance of fitness and healthy body weight, policy requirements are uneven: in a survey of 401 chiefs, nearly all emphasized fitness, yet a majority of departments lacked maintenance mandates (Wagner et al., 2023). Reported concerns about obesity included functional limits (e.g., stamina, self-protection) and potential loss of public confidence–channels that map directly onto perceived competence and agency. These organizational beliefs underscore why body composition belongs alongside race and gender in experiments on officer perception.

In sum, AI-generated imagery provides a reproducible means to manipulate race, gender, and body composition while holding other features constant. Our validation tests whether these synthetic profiles transmit intended attributes and whether observers register the manipulations as theoretically meaningful differences.

Significance of this study

This study advances experimental social science by validating a scalable method to create systematically varied, reproducible human stimuli. Across politics, organizations, and the military, appearance-based judgments forecast who attains leadership; in policing, similar perceptual processes plausibly shape both public evaluations and

[2] The criminological tradition has long attended to physique and social evaluation (Sheldon et al., 1940; Sheldon, 1949; Maddan et al., 2008; McCandless et al., 1972).

the internal reproduction of organizational elites. Yet much evidence to date relies on small, idiosyncratic photo sets or observational designs that limit causal inference and replication. By validating AI-generated profiles that reliably transmit intended attributes (race, gender, body type) and pass manipulation checks, we provide an identification-ready stimulus toolkit that generalizes beyond policing to any domain where visual cues are theoretically central.

Methodologically, the contribution is threefold. First, we establish construct validity for AI-generated human images: participants recognize intended traits and detect targeted changes across matched sets. Second, we implement stimulus sampling (multiple exemplars per cell) to avoid idiosyncratic-image artifacts, improving generalization over stimuli and supporting cumulative science. Third, we document a transparent generation protocol that others can reuse, enabling preregistration of treatments, precise replication, and multi-site pooling–features rarely feasible with archival photographs or bespoke imagery.

Substantively, the framework enables cleaner tests of how embodied cues structure core dimensions of social perception such as competence, agency, and warmth. More broadly, validated synthetic stimuli open avenues for comparative and cross-domain experiments (e.g., politics, workplaces, courts) that isolate visual mechanisms under controlled conditions. In short, we move the field from constrained, non-replicable imagery toward a reproducible, scalable, and causally informative approach to studying how visual information organizes social judgment.

As recent work in Science and PNAS observes, the promise of generative AI for social science lies in producing scalable, realistic stimuli–but only if researchers can ensure validity, transparency, and replicability (Grossmann et al., 2023; Bail, 2024; Nature Human Behaviour Editorial Board, 2023). This study reflects on this challenge, providing evidence that AI-generated human profiles reliably convey intended traits and elicit predictable patterns of social judgment.

Methods

Research design and procedure

This study uses original data from an online survey experiment, conducted in September 2025, designed to assess whether participants can accurately identify the intended attributes (race, gender, and body type) of police officer profiles in AI-generated images. The experiment employs a $4 \times 2 \times 3$ factorial design, systematically varying the visual stimuli along race, gender, and body-composition, resulting in 24 unique treatment cells (see Table 1). Following Judd et al. (2017), our experiment is a CNC design (participants crossed with condition; targets nested within condition; participants and targets crossed). Accordingly, we fit mixed-effects models with random intercepts for targets and random intercepts/slopes for participants, estimating variance components with crossed random effects for participants and targets.

To enhance the robustness of our method, we implemented a stimulus sampling strategy (Wells & Windschitl, 1999; Judd et al., 2012; Monin & Oppenheimer, 2014) to mitigate potential idiosyncratic effects tied to any single image. Within

Table 1 Experimental design factors

Factor	Levels
Race	White
	Black
	Asian
	Latino
Gender	Male
	Female
Somatotype	Ectomorph (slim, low body fat)
	Mesomorph (muscular, athletic)
	Endomorph (larger rounded body shape, high body fat)

Table 2 Generic image-generation prompt with brackets marking manipulated traits

Example Image-Generation Prompt
Produce a professional police officer headshot photograph (head to waist) for experimental research. The officer is [RACE], [GENDER], and has a [BODY TYPE] build. The officer wears a standard U.S. police uniform with neutral facial expression, plain background, even lighting, and realistic photographic style. No accessories, no text, no background objects.

each treatment cell (e.g., White male mesomorph), we generated multiple distinct AI officer profiles using structured ChatGPT 4o and eventually ChatGPT 5 prompts. While the prompts consistently specified race, gender, and body type, we introduced minor variations to produce perceptually distinct but trait-consistent exemplars. This approach reduces the likelihood that results are driven by peculiarities of individual images rather than the intended experimental manipulation. We include a generation parameter example to support transparency and facilitate replication (see Table 2).

We recruited 513 participants via Prolific to ensure a diverse and representative U.S.-based sample, and we report the full sample's descriptive results in Table 3. We derived this sample size from a simulation based power analysis. Eligibility criteria included being at least 18 years old, fluent in English, and residing in the United States. Each participant was presented with a randomized subset of the images–ensuring that no respondent viewed multiple exemplars from the same treatment cell–to avoid intra-condition carryover effects. Participants were compensated for their time.

Image generation and trait manipulation

The AI-generated officer profiles were designed to closely resemble real-world police headshots in uniform, background, and lighting. We used precise LLM prompts to create a specific combination of race, gender, and body type which we adapt from Sheldon et al. (1940)'s concept of somatotype, with image quality and realism verified through internal checks before inclusion[3]

[3] Minor adjustments were sometimes necessary after image generation. For example, when an image did not properly reflect the intended trait, we applied a reprompt to correct it. In addition, all backgrounds were standardized using photo-editing software, and badges or nameplates were covered to reduce potential sources of bias.

Table 3 Respondent descriptives

Variable	Category	N	%
Gender	Female	257	50.1
	Male	254	49.5
	Other	2	0.4
Race/Ethnicity	American Indian or Alaskan Native	7	1.4
	Asian	35	6.8
	Black or African American	65	12.7
	Hispanic or Latino	55	10.7
	Middle Eastern or North African	3	0.6
	Native Hawaiian or Pacific Islander	1	0.2
	Other	18	3.5
	White	329	64.1
Partisanship	Democrat	196	38.2
	Independent	159	31.0
	Republican	158	30.8
Education	Less than high school degree	2	0.4
	High school graduate (high-school diploma or GED)	72	14.0
	Some college, but no degree	103	20.1
	Associate's degree (two-year)	66	12.9
	Bachelor's degree (four-year)	199	38.8
	Master's degree	57	11.1
	Professional degree (JD, MD, etc.)	8	1.6
	Doctoral degree (PhD)	6	1.2
Age	18-29	97	18.9
	30-44	142	27.7
	45-64	210	40.9
	65 and over	64	12.5
Income	Less than $20,000	51	9.9
	$20,000-$39,999	89	17.3
	$40,000-$59,999	103	20.1
	$60,000-$79,999	75	14.6
	$80,000-$99,999	69	13.5
	$100,000-$149,999	78	15.2
	More than $150,000	48	9.4

Observable traits coding (Task 1)

To validate whether the generated images reliably conveyed the intended traits, participants were asked to classify a random subset of the officer images across the following dimensions:

- **Race Identification:** "What race does this officer appear to be?" (White, Black, Asian, Hispanic)

- **Gender Identification:** "What gender does this officer appear to be?" (Male, Female)
- **Somatotype Identification:** "How would you describe this officer's body type?" (Ectomorph, Mesomorph, Endomorph)

Responses were recorded using a standardized multiple-choice format. Each image was independently rated by multiple participants to allow for assessment of inter-rater consistency. We computed inter-rater reliability using Krippendorf's α and percentage agreement.

To measure confoundedness across trait dimensions, we conducted a Principal Components Analysis (PCA) with direct oblimin rotation on the trait responses (Adams et al., 2024). This analysis tested the empirical independence of race, gender, and somatotype classifications (see Appendix A.2).

Trait recognition hypotheses (Task 1)

- **H1a**: Participants will accurately identify the race of AI-generated officer profiles at rates significantly above chance.
- **H1b**: Participants will accurately identify the gender of AI-generated officer profiles at rates significantly above chance.
- **H1c**: Participants will accurately identify the body type (somatotype) of AI-generated officer profiles at rates significantly above chance.

Observable traits change (Task 2)

The second task asked participants to compare sets of officer images presented side by side. Within each set, one officer differed on a single manipulated trait while all other features remained constant. Respondents were asked to identify which trait differed, with options including race or ethnicity, gender, body type, facial expression, uniform, or age. This task provided a direct test of whether participants could perceive intended manipulations across otherwise identical profiles. Importantly, the number of observations (N) is not constant across comparison types because the factorial design generated unequal numbers of available image sets for each trait. For example, there were more unique race and body type contrasts than gender contrasts, which produced a higher N for those dimensions. In addition, random assignment of image sets to participants meant that not every respondent evaluated every type of change, further contributing to variation in the total number of observations by trait.

Trait change hypotheses (Task 2)

- **H2a**: Participants will accurately identify changes in officer race or ethnicity at rates significantly above chance.
- **H2b**: Participants will accurately identify changes in officer gender at rates significantly above chance.
- **H2c**: Participants will accurately identify changes in officer body type (somatotype) at rates significantly above chance.

Officer impressions (Task 3)

Building on Carrier et al. (2014); Brambilla et al. (2011), who disentangled agency from competence by restricting agency to its motivational component, we adopted validated short scales for competency, agency, and warmth. In the final task, respondents evaluated each officer on a set of impression measures that capture three fundamental dimensions of social judgment: agency, competence, and warmth. Respondents rated officers on nine trait adjectives using a seven-point scale ranging from 1 (Not at all) to 7 (Extremely). Competence was measured with competent, capable, and efficient. Agency was measured with ambitious, self-confident, and assertive. Warmth was measured with warm, friendly, and likable. This structure allows us to separately assess whether officers convey the motivation to advance the self (agency), the ability to effectively pursue goals (competence), and concern for others (warmth).

We independently assessed the internal consistency of the three impression scales using McDonald's ω, which provides a more general reliability estimate than Cronbach's α (McNeish, 2018). Reliability was excellent across all three scales: competence (ω = .95), agency (ω = .90), and warmth (ω = .96). These values indicate that the items within each scale cohered strongly and justify treating them as composite measures of the intended constructs (see Table 4).

Impressions hypotheses (Task 3)

- **H3a**: Participants will rate male officers as more competent and agentic, but less warm than female officers (Mize, 2025; Mize & Manago, 2018; Ebert et al., 2014)
- **H3b**: Participants will rate endomorphic officers as less competent and agentic than ectomorphic officers (Levine & Schweitzer, 2015; Puhl & Brownell, 2013, 2006; Roehling, 1999), but there will be no effect for warmth (Levine & Schweitzer, 2015)
- **H3c**: Participants will rate mesomorphic officers as more competent, agentic, and warm than ectomorphic officers. (Lukaszewski et al., 2016; Zeng et al., 2022) **H3d**: Partisipants will rate White and Asian officers as more competent than latino or black officers (Heiserman, 2023), but White officers will be rated as less warm (Heiserman, 2023; Levin & Thomas, 1997; Dunbar et al., 2024; Pica et al., 2020)

Results

Observable traits agreement

The first set of analyzes tested the trait recognition hypotheses (H1a-H1c). Table 5 reports the proportion of participants who correctly classified each officer profile's

Table 4 Impression rating descriptives

Scale	N	Mean	SD	Min	Max
Competence	2565	4.98	1.29	1	7
Agency	2565	4.88	1.30	1	7
Warmth	2565	4.35	1.40	1	7

Table 5 Agreement between true and perceived traits

Trait	N	Correct	Percent Agreement	By-chance p	Krippendorff's α
Female Officers					
Gender	1326	1230	92.8	0	
Body Type	1326	1082	81.6	0	
Race	1326	1151	86.8	0	
Male Officers					
Gender	1239	1233	99.5	0	
Body Type	1239	1099	88.7	0	
Race	1239	1168	94.3	0	
Overall					
Gender	2565	2463	96.0	0	0.872
Body Type	2565	2181	85.0	0	0.718
Race	2565	2319	90.4	0	0.816

race, gender, and body type, disaggregated by the true[4] gender[5] of the officer image. Across all traits, agreement between the intended (true) characteristics and participant classifications was consistently high, with by-chance tests indicating performance significantly above random guessing (all $p < 0.001$).

Participants were especially accurate in identifying officer gender, with agreement rates exceeding 92% for female profiles and 99% for male profiles. This strong performance provides clear support for H1b, demonstrating that the AI-generated profiles reliably conveyed gender. Similarly, officer body type was accurately perceived, with agreement rates of 81.7% for female officers and 88.6% for male officers, strongly supporting H1c. Finally, classification of race was also well above chance, with agreement rates of 86.8% for female officers and 94.2% for male officers, consistent with H1a. Here, we demonstrate that the AI-generated officer profiles conveyed the intended demographic and somatotype characteristics with high fidelity. Additionally, these findings provide the necessary foundation for subsequent tasks: assessing differentiation across manipulated traits (Task 2) and evaluating impressions of competence, agency, and warmth (Task 3). Additionally, we find that results slightly improve upon attention check implementation (see Appendix A.5)

In addition to percent agreement, we assessed inter-rater reliability using Krippendorff's α, which accounts for chance agreement, multiple raters, and missing data (Krippendorff, 2019). Reliability was high for gender (α = .87), strong for race (α = .82), and moderate for body type (α = .72). These values indicate that respondents not only performed above chance but also reached satisfactory levels of consistency in their classifications, particularly for gender and race. According to Krippendorff (2019), our α for body type is within the range for the lower bound for tentative conclusions.

Mixed-effects models of trait recognition To further evaluate the perceptual validity of the AI-generated officer profiles, we estimated mixed-effects logistic regression models (Judd et al., 2017) predicting correct identification of gender, race, and body

[4] "True" characteristics are determined by the LLM prompt used to generate the image

[5] See Table 9 for accuracy by officer race and Table 10 for accuracy by officer body type

type (see Appendix A.5.3). Each model included fixed effects for the officer's true race, gender, and body type, with random intercepts for both respondents and profiles.

Baseline accuracy was very high across outcomes (intercepts: gender β = 16.149, p < 0.001; race β = 4.883, p < 0.001; body type β = 3.018, p <0.001). Female officers were significantly less likely to be correctly identified than male officers for gender (β = -4.447, p < 0.001) and race (β = -0.989, p = 0.008), with no reliable difference for body type (β = -0.569, p = 0.150). For race recognition, respondents were more accurate for Black officers (β = 1.903, p = 0.007), but less accurate for Asian officers (β = -1.538, p = 0.002) and Latino officers (β = -1.798, p < 0.001). For body-type recognition, mesomorphic officers were less likely to be correctly recognized than ectomorphs (β = -1.298, p = 0.006), whereas endomorphic officers were more likely (β = 1.088, p = 0.028).

Trait change agreement

The second set of analyses tested the trait-change recognition hypotheses (H2a - H2c). Table 6 reports results from the trait change task (task 2), where participants were presented with sets of officer profiles in which only one characteristic (race, gender, or body type) was systematically varied. Respondents were asked to identify which trait differed across the images. Across all categories, accuracy was substantially higher than chance, with by-chance tests confirming reliable detection of intended manipulations (all p < 0.001).

Notably, participants were comparably effective at detecting changes across the three focal traits. Agreement rates were 71.1% for body type, 71.4% for gender, and 70.9% for race or ethnicity. This shows clear support for the differentiation component of our validation, demonstrating that respondents could reliably perceive when an officer's profile differed on a targeted attribute.

Extending the evidence from Section "Observable traits agreement". Not only did participants accurately recognize the intended traits of individual images, they also consistently detected manipulated changes across otherwise identical profiles. This reinforces the conclusion that the AI-generated officer profiles systematically convey the intended demographic and physical characteristics and that participants perceive these manipulations as meaningful and distinct. In other words, we show that AI generated images can precisely generate certain traits while controlling for others. Again, we find that results are robust and slightly improve when we implement attention check analyses (see Appendix A.5.2).

Although Krippendorff's α is a useful index of inter-rater reliability in many content-analysis settings, it is not the best measure for the change-detection task in Study 2. This task differs fundamentally from a coding exercise in which multiple raters independently classify the same units. Instead, each respondent was presented with a unique set of triads and asked to identify which trait changed. Because the task

Table 6 Agreement between true change and perceived change

Trait Change	N	Correct	Respondent Accuracy	By-chance p
Gender	311	222	71.4%	0
Body Type	1320	938	71.1%	0
Race or ethnicity	934	662	70.9%	0

has an objectively correct answer, accuracy relative to chance is the more direct test of our hypotheses. In this context, α is misleading: it penalizes variability in the specific incorrect categories chosen by respondents, even when overall accuracy is well above chance. Thus, while α estimates were low (.23), this does not indicate poor construct validity. Rather, it reflects that when participants erred, they did not consistently select the same wrong category. For this reason, we rely on percent agreement relative to chance as the most appropriate indicator of performance in this task.

Impressions: main effects

The third set of analyses tested the impressions hypotheses (H3a–H3d) by examining whether respondents evaluated officers differently on competence, agency, and warmth as a function of race, gender, and body type. Table 7 presents the main effects from respondent and image random-effects models, and Figs. 1 and 2 display conditional means and average effect sizes for each trait dimension.

Gender. Consistent with the direction of H3a but not statistically significant, female officers were rated lower than male officers on competence (-0.038, n.s.) and agency (-0.108, n.s.). At the same time, female officers were evaluated as warmer (+0.173, $p < 0.01$). The pattern aligns with established gender stereotyping: female officers are perceived as warmer and, directionally, slightly less competent and agentic, indicating that AI-generated profiles elicit theoretically coherent evaluations even though only warmth reached significance.

Table 7 Key outcomes

	Competence	Agency	Warmth
Intercept	5.017***	4.917***	3.999***
	(0.093)	(0.102)	(0.111)
Black officer	0.067	0.113	0.272*
	(0.091)	(0.103)	(0.109)
Latino officer	0.051	0.038	0.220*
	(0.091)	(0.103)	(0.109)
Asian officer	0.156	-0.012	0.312**
	(0.090)	(0.103)	(0.109)
Female officer	-0.038	-0.108	0.173*
	(0.064)	(0.073)	(0.077)
Mesomorph	0.376***	0.555***	0.215*
	(0.079)	(0.089)	(0.094)
Endomorph	-0.625***	-0.608***	-0.037
	(0.078)	(0.089)	(0.094)
Num.Obs.	2565	2565	2565
R2 Marg.	0.105	0.134	0.017
R2 Cond.	0.617	0.572	0.595
ICC	0.6	0.5	0.6
RMSE	0.72	0.77	0.80
Random intercept: respondent	Yes	Yes	Yes
Random intercept: profile	Yes	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Reference categories: White, Male, Ectomorph. Models include random intercepts for respondent and profile

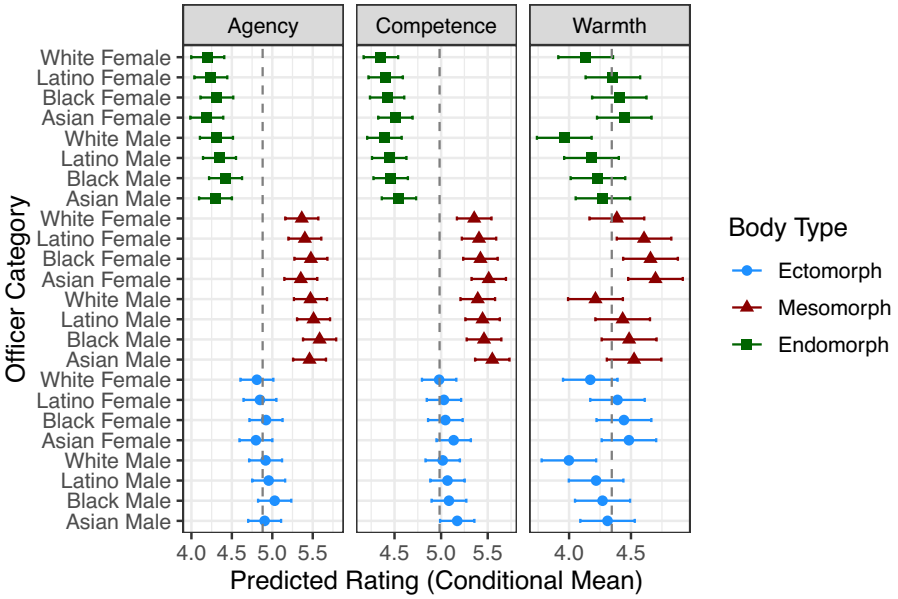


Fig. 1 Predicted impression ratings by officer category across the three dimensions of social perception. Agency, Competence, and Warmth are shown for officers crossed by race and gender, with separate trends by somatotype (ectomorph, mesomorph, endomorph). Points are conditional means from mixed-effects models with random intercepts for respondent and profile; error bars denote 95% confidence intervals. Ratings use a 1–7 scale (1 = not at all, 7 = extremely). Sample size: $N = 2{,}565$ profile ratings. Numerical axes reflect the measurement scale (1–7). A dashed grey vertical line indicates the mean for each panel

Race. Relative to White officers (reference), Black and Latino officers did not differ significantly on competence or agency (competence: 0.067 and 0.051; agency: 0.113 and 0.038; all n.s.). Asian officers were also similar to Whites on competence and agency (competence: 0.156, n.s.; agency: -0.012, n.s.). In contrast, warmth ratings were consistently higher for minority officers: Black (0.272, $p < .05$), Latino (0.220, $p < .05$), and Asian (0.312, $p < .01$). These results provide support for the warmth component of H3d, while competence and agency show no reliable racial differences.

Body type. Body type produced some of the largest and most consistent effects across outcomes, in line with H3b and H3c. Relative to ectomorphic officers (the reference category), mesomorphic officers were rated substantially higher on competence (+0.376, $p < .001$), agency (+0.555, $p < .001$), and warmth (+0.215, $p < .05$). In contrast, endomorphic officers were penalized on competence (-0.625, $p < .001$) and agency (-0.608, $p < .001$), but showed no significant difference in warmth (-0.037, n.s.). These results replicate longstanding stereotypes that reward muscular body types with perceptions of capability and motivation, while heavier builds face sharp penalties on status-linked traits. Importantly, warmth evaluations were largely unaffected by endomorphic profiles, highlighting that penalties were specific to competence and agency rather than extending to interpersonal likability.

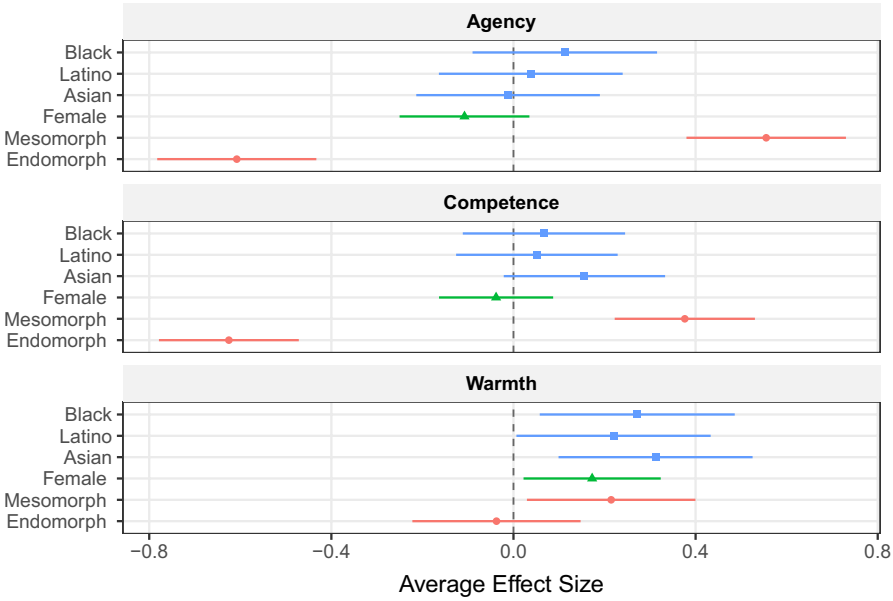


Fig. 2 Average marginal effects of officer attributes on impression ratings relative to reference categories (White, Male, Ectomorph). Agency, Competence, and Warmth panels display coefficient estimates with 95% confidence intervals from mixed-effects models that include random intercepts for respondent and profile. Positive values indicate higher perceived Agency/Competence/Warmth relative to the reference; negative values indicate lower perceptions. Sample size: $N = 2{,}565$ profile ratings. Numerical axes are centered on zero to show direction and magnitude of effects (zero = no difference)

Overall patterns. Figure 1 displays conditional means across officer categories, highlighting the strong role of body type in shaping evaluations. Mesomorphic officers consistently received the most favorable ratings, while endomorphic officers faced the steepest penalties on competence and agency. In contrast, racial differences were modest on competence and agency but pronounced on warmth, with all minority groups rated significantly warmer than White officers. Gender effects followed a similar pattern: women were rated warmer, with small and nonsignificant disadvantages on competence and agency. Figure 2 summarizes these patterns by effect size, showing that body type exerted the largest influence on competence and agency, whereas race and gender were more consequential for warmth. Notably, white male officers (who comprise the modal demographic category in U.S. policing (Wilson & Grammich, 2022)) are perceived as among the least warm and do not receive compensatory advantages on competence or agency. Situated within the policing literature, this pattern suggests that demographic prototypicality within law enforcement does not translate into more favorable public evaluations and may instead be associated with perceptions tied to institutional authority and adversarial relations rather (Fiske, 2018; Fiske et al., 2002) than interpersonal approachability. Together, these results demonstrate that respondents systematically evaluate AI-produced officer profiles along theoretically coherent dimensions, with body

type driving perceptions of capability and race and gender shaping interpersonal judgments. Notably, all results are robust to attention check analyses (see Appendix A.5.3).

Limitations and future research

Although the study demonstrates the value of AI-generated officer profiles, several limitations should be recognized. Most importantly, the study relied on a U.S.-based sample, which may limit the generalizability of results to other cultural or national contexts where perceptions of police differ, or where different normative ideals of sex, race, and body types may hold. Additionally, the current design focused on three core attributes (race, gender, and body type) while other potentially important characteristics such as age, facial expression, or uniform details remain untested. While our design is robust to unobserved confounders, those same confounds may provide additional explanatory power and can likely serve as additional focal concerns in work building on this method.

Future research can build on this study in several ways. First, expanding the set of validated traits to include age, facial expression, rank, and uniform features would provide an even broader toolkit for studying perceptions of police. Future work will also extend this framework to additional embodied and appearance-based cues that are increasingly salient in contemporary policing, including facial hair, glasses, hair loss, visible tattoos, facial morphology (e.g., jawline and symmetry), relative height, and variation at the upper end of the body-size distribution. The generative approach validated here is well suited to systematically manipulating these features–both independently and in combination–allowing researchers to examine how such cues shape public and organizational judgments under controlled conditions. Second, extending validation efforts started by Wu et al. (2026) to dynamic formats such as AI-generated video could allow researchers to examine how traits interact with movement, speech, or behavior. Third, combining officer profiles with situational contexts such as narratives of encounters or trial transcripts would allow for richer tests of how individual and contextual cues interact in shaping perceptions of law enforcement. Finally, replication across diverse populations and cross-national settings would strengthen confidence in the generalizability of these findings and support comparative studies of policing and social perception.

Conclusion

This study addresses two linked problems in experimental social science: the scarcity of reproducible visual stimuli and the systematic neglect of *body composition* as a first-order visual cue. Building on evidence that appearance-based judgments shape leadership selection across institutions (Todorov et al., 2005; Antonakis & Dalgas, 2009; Berggren et al., 2010; Laustsen, 2014; Olivola & Todorov, 2010a, b; Mueller & Mazur, 1996) and the policing application that connects thin-slice

facial judgments to real promotion outcomes (Adams et al., 2024), we develop and validate AI-generated officer profiles that permit controlled manipulation of race, gender, and somatotype.

Empirically, three results matter. First, the synthetic profiles exhibit strong construct validity: observers reliably identify intended traits. Second, observers detect targeted changes across matched images, indicating that manipulations register as theoretically meaningful differences rather than idiosyncratic artifacts. Third, impressions along competence, agency, and warmth reproduce well-established patterns from social perception and status research, including gender differences and sizable penalties/advantages tied to body composition (Ebert et al., 2014; Levine & Schweitzer, 2015; Lukaszewski et al., 2016; Mize, 2025; Mize & Manago, 2018; Puhl & Brownell, 2013, 2006; Roehling, 1999; Zeng et al., 2022). Together with stimulus sampling (multiple exemplars per cell), these findings show that AI-generated imagery can deliver identification-ready treatments while generalizing over stimuli.

Substantively, treating body composition as co-equal with race and gender clarifies a mechanism often sidelined in prior work: cues of weight and formidability function as status signals with downstream implications for who is seen as competent, agentic, and ultimately leadership material (Blaker & van Vugt, 2014; Lukaszewski et al., 2016; Zeng et al., 2022). In policing, where leaders set policy, hiring, and discipline, such perceptual inputs can scale into institutional consequences (Adams et al., 2024). Methodologically, the contribution is a transparent, reusable protocol that shifts the field from observational forecasting with archival photos to causal tests using reproducible, shareable visual treatments.

The practical takeaways are straightforward: (i) validated synthetic images let researchers isolate visual mechanisms with high control; (ii) body composition should be modeled as a core attribute; and (iii) stimulus sampling is essential for credible generalization beyond any single image. Ultimately, the approach is portable to other domains where previous work has demonstrated visual judgments matter, including elections, workplaces, courts. In sum, validated, systematically varied synthetic profiles provide a scalable foundation for experiments on how visual information organizes social judgment across institutions.

Appendix items

A.1 Survey instrument

[6]

- **Demographics**Gender: Male, Female, Other

[6] Tasks are in reverse order from the main text description as to not cue participants on design of study

- Race/Ethnicity: White, Black or African American, Hispanic or Latino, Asian, American Indian or Alaskan Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander, Other
- Age: 18–29, 30–44, 45–64, 65+
- Education: Less than high school, High school graduate, Some college (no degree), Associate's degree, Bachelor's degree, Master's degree, Professional degree (JD, MD), Doctoral degree (PhD)
- Partisanship: Democrat, Republican, Independent
- Income: Less than $20,000; $20,000–39,999; $40,000–59,999; $60,000–79,999; $80,000–99,999; $100,000–149,999; More than $150,000**Officer Impressions (Task 1)**

Participants viewed a single officer image at a time and rated how well the following traits described the officer on a 7-point scale (1 = Not at all, 7 = Extremely):

- Competent, Capable, Efficient
- Ambitious, Self-Confident, Assertive
- Warm, Friendly, Likeable**Observable Traits Coding (Task 2)**

Participants answered:

- "What race or ethnicity does this individual appear to be?" (White, Black, Latino/Hispanic, Asian)
- "What gender does this individual appear to be?" (Male, Female, Other)
- "How would you describe this individual's body type?" (Slim/narrow, Muscular/athletic, Rounder body/higher body fat)
- "Which age group does this individual appear to belong to?" (Under 26, 26–34, 35+)**Observable Traits Change (Task 3)**

Participants were shown sets of officer images side by side, identical except for one trait. They were asked:

"Which trait is most distinct across the officers?"

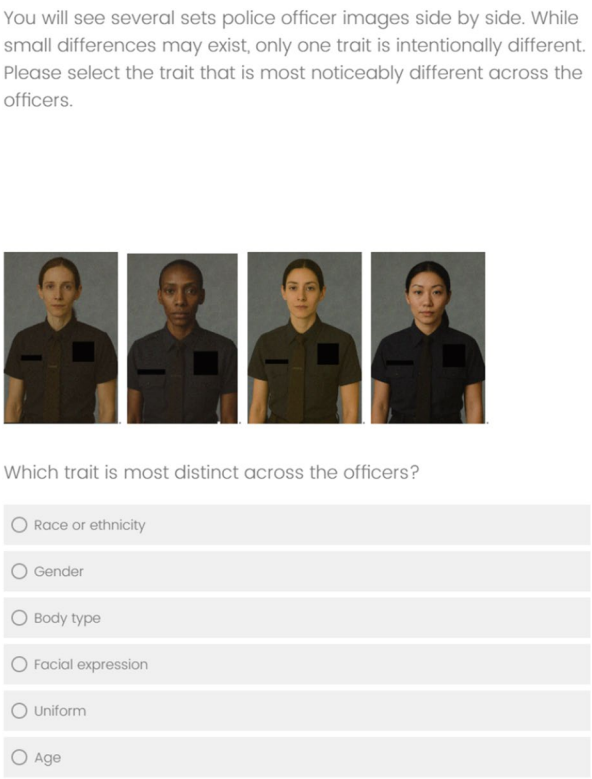
Response options:

- Race or ethnicity
- Gender
- Body type
- Facial expression
- Uniform
- Age

A.2 Task 1

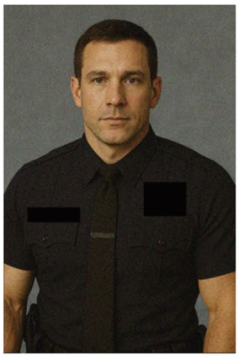
Principal components analysis with direct oblimin rotation produced a three-factor solution. Race loaded cleanly on a single factor (loading = .91), while gender and body type showed moderate cross-loadings. This indicates that the two traits were generally distinguishable but not fully independent: when respondents misclassified one, they were somewhat more likely to misclassify the other.

A.3 Task 2



Example of the trait change identification task (Task 2). Respondents viewed sets of officer images displayed side by side. One image differs on a single manipulated trait (race, gender, or body type), while all other features remain constant. Respondents were asked to identify which trait had changed. This task directly tests whether participants perceive intended manipulations across otherwise identical profiles. Each set was shown to multiple independent respondents (*N* varies by trait type; see text)

A.4 Task 3



Based only on this image, how much do you think each of the following words describes this officer?

	1 (Not at all)	2	3	4	5	6	7 (Extremely)
Competent	○	○	○	○	○	○	○
Capable	○	○	○	○	○	○	○
Efficient	○	○	○	○	○	○	○
Ambitious	○	○	○	○	○	○	○
Self-Confident	○	○	○	○	○	○	○
Assertive	○	○	○	○	○	○	○
Warm	○	○	○	○	○	○	○
Friendly	○	○	○	○	○	○	○
Likeable	○	○	○	○	○	○	○

Example of the impression-rating task (Task 3). Respondents evaluated each officer profile on three dimensions of social perception: competence, agency, and warmth. Participants rated nine trait adjectives (three per construct) using a 1–7 scale (1 = not at all, 7 = extremely). Reliability of the composite scales was high (Competence $\omega = .95$, Agency $\omega = .90$, Warmth $\omega = .96$)

A.5 Results

A.5.1 Observable traits agreement

Table 8 Agreement between true and perceived traits: attention check

Trait	N	Correct	Percent Agreement	By-chance p	Krippendorff's α
Female Officers					
Gender	1267	1177	92.9	0	
Body Type	1267	1036	81.8	0	
Race	1267	1103	87.1	0	
Male Officers					
Gender	1178	1172	99.5	0	
Body Type	1178	1050	89.1	0	
Race	1178	1111	94.3	0	
Overall					
Gender	2445	2349	96.1	0	0.874
Body Type	2445	2086	85.3	0	0.726
Race	2445	2214	90.6	0	0.820

Table 9 Agreement between true and perceived traits: by race

Officer Race	N	Gender Accuracy	Race Accuracy	Body Type Accuracy
White	703	96.7%	95.3%	84.6%
Black	651	94.6%	99.1%	90.2%
Asian	632	97.6%	81.8%	81.0%
Latino	579	95.0%	84.1%	84.1%

Table 10 Agreement between true and perceived traits: by body type

Officer Body Type	N	Gender Accuracy	Race Accuracy	Body Type Accuracy
Ectomorph	863	96.6%	91.8%	87.3%
Mesomorph	823	96.4%	92.8%	72.1%
Endomorph	879	95.1%	86.8%	95.0%

A.5.2 Trait change agreement

Table 11 Trait-change recognition accuracy by manipulated attribute

Trait Change	N	Correct	Respondent Accuracy	By-chance p
Race or ethnicity	889	640	72.0%	0
Body Type	1261	907	71.9%	0
Gender	295	212	71.9%	0

A.5.3 Impressions

Table 12 Recognition hererogeneity

	Gender recognition	Race recognition	Body type recognition
Intercept	16.149***	4.883***	3.018***
	(2.186)	(0.560)	(0.517)
Female	-4.447***	-0.989**	-0.569
	(1.050)	(0.379)	(0.392)
Black	-0.546	1.903**	0.296
	(1.201)	(0.680)	(0.550)
Asian	0.998	-1.538**	-0.069
	(1.303)	(0.504)	(0.555)
Latino	-0.327	-1.798***	-0.195
	(1.293)	(0.503)	(0.546)
Mesomorph	-0.961	0.031	-1.298**
	(1.126)	(0.471)	(0.466)
Endomorph	-1.256	-0.747	1.088*
	(1.143)	(0.455)	(0.501)
Num.Obs.	2565	2565	2565
R2 Marg.	0.067	0.303	0.157
R2 Cond.	0.960	0.616	0.517
ICC	1.0	0.4	0.4
RMSE	0.09	0.22	0.28
Random intercept: respondent	Yes	Yes	Yes
Random intercept: profile	Yes	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Reference categories: White, Male, Ectomorph. Models include random intercepts for respondent and profile

Table 13 Key outcomes: attention check

	Competence	Agency	Warmth
Intercept	5.010***	4.920***	3.981***
	(0.096)	(0.104)	(0.113)
Black officer	0.080	0.128	0.254*
	(0.094)	(0.105)	(0.111)
Latino officer	0.051	0.041	0.200
	(0.093)	(0.104)	(0.111)
Asian officer	0.174	0.004	0.288**
	(0.093)	(0.104)	(0.111)
Female officer	-0.034	-0.106	0.171*
	(0.066)	(0.074)	(0.078)
Mesomorph	0.371***	0.540***	0.239*
	(0.081)	(0.091)	(0.096)
Endomorph	-0.651***	-0.628***	-0.019
	(0.081)	(0.090)	(0.096)
Num.Obs.	2445	2445	2445
R2 Marg.	0.108	0.134	0.017
R2 Cond.	0.622	0.577	0.599
ICC	0.6	0.5	0.6
RMSE	0.72	0.77	0.80
Random intercept: respondent	Yes	Yes	Yes
Random intercept: profile	Yes	Yes	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Reference categories: White, Male, Ectomorph. Models include random intercepts for respondent and profile

Data, Materials, and Code Availability The datasets, materials, and code needed to replicate this study are available in [https://github.com/cgrob50/AI-Experimental-Stimuli].

Declarations

Competing Interest The authors have no relevant financial or non-financial interests to disclose. Replication materials will be made available upon publication. This study is intended to be considered as an original research article.

References

Adams, I. T., Mourtgos, S. M., Simon, C. A., & Lovrich, N. P. (2024). If the face fits: predicting future promotions from police cadets' facial traits. *Journal of experimental criminology, 20*(3), 937–972.

Antonakis, J., & Dalgas, O. (2009). Predicting elections: Child's play! *Science, 323*(5918), 1183–1183.

Bail, C. A. (2024). Can Generative AI improve social science? *Proceedings of the National Academy of Sciences, 121*(21), Article e2314021121.

Berggren, N., Jordahl, H., & Poutvaara, P. (2010). The looks of a winner: Beauty and electoral success. *Journal of public economics, 94*(1–2), 8–15.

Berggren, N., Jordahl, H., & Poutvaara, P. (2017). The right look: Conservative politicians look better and voters reward it. *Journal of Public Economics, 146*, 79–86.

Bird, J. J., & Lotfi, A. (2024). CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images. *IEEE Access, 12*, 15642–15650.

Blaker, Nancy M. and Mark van Vugt. 2014. The Status-Size Hypothesis: How Cues of Physical Size and Social Status Influence Each Other. In *The Psychology of Social Status*, ed. Joey Cheng, Jessica L. Tracy and Cameron Anderson. New York, NY: Springer pp. 119–137.

Bordo, S. (1993). *Unbearable Weight: Feminism, Western Culture, and the Body*. Berkeley, CA: University of California Press.

Brambilla, M., Rusconi, P., Sacchi, S., & Cherubini, P. (2011). Looking for honesty: The primary role of morality (vs. sociability and competence) in information gathering. *European Journal of Social Psychology, 41*(2), 135–143.

Carrier, A., Louvet, E., Chauvin, B., & Rohmer, O. (2014). The Primacy of Agency over Competence in Status Perception. *Social Psychology, 45*(5), 347–356.

Castaneda, G. 2018. Public Perception of Police Behavior During a Traffic Stop Doctoral dissertation Alliant International University. Publication no. 10935723.

Cooke, Di, Abigail Edwards, Sophia Barkoff and Kathryn Kelly. 2025. "As Good As A Coin Toss: Human detection of AI-generated images, videos, audio, and audiovisual stimuli.". https://arxiv.org/abs/2403.16760

Duguid, Michelle M. and Jack A. Goncalo. 2011. "Living Large: The Powerful Overestimate Their Own Height." *Psychological Science* 23(1):36–40. Original work published 2012. https://doi.org/10.1177/0956797611422915

Dunbar, A., P. A. Hanink and B. Kyle. 2024. "Good Cop, Bad Cop, Black Cop, White Cop: Unraveling Public Perceptions of Police Use of Force." *Crime & Delinquency* pp. 1–23.

Ebert, I. D., Steffens, M. C., & Kroth, A. (2014). Warm, but maybe not so competent?—Contemporary implicit stereotypes of women and men in Germany. *Sex roles, 70*(9), 359–375.

Epstein, David C., Ishan Jain, Oliver Wang and Richard Zhang. 2023. Online Detection of AI-Generated Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*. pp. 382–392.

Ewanation, L., & Maeder, E. M. (2023). The influence of race on jurors' perceptions of lethal police use of force. *Law and Human Behavior, 47*(1), 53–67.

Fiske, S. T. (2018). Stereotype Content: Warmth and Competence Endure. *Current Directions in Psychological Science, 27*(2), 67–73.

Fiske, S. T., Cuddy, A. J. C., Glick, P., & Jun, X. (2002). A Model of (Often Mixed) Stereotype Content: Competence and Warmth Respectively Follow from Perceived Status and Competition. *Journal of Personality and Social Psychology, 82*(6), 878–902.

Gordon, R. A., Crosnoe, R., & Wang, X. (2013). Physical Attractiveness and the Accumulation of Social and Human Capital in Adolescence and Young Adulthood: Assets and Distractions. *Monographs of the Society for Research in Child Development, 78*(6), 1–137.

Grossmann, I., Feinberg, M., Parker, D. C., Christakis, N., Tetlock, P. E., & Cunningham, W. A. (2023). AI and the transformation of social science research. *Science, 380*(6650), 1108–1109.

Heiserman, N. (2023). Intersectional Complexity in Stereotype Content. *American Journal of Sociology, 129*(3), 753–795. https://doi.org/10.1086/728051

Holbrook, Colin and Daniel M. T. Fessler. 2015. The same, only different: Threat management systems as homologues in the tree of life. In *Handbook of Personal Security*, ed. Patrick J. Carroll, Robert M. Arkin and Alana L. Wichman. New York, NY: Psychology Press pp. 95–109.

Hwang, H. C., & Matsumoto, D. (2014). Dominance threat display for victory and achievement in competition context. *Motivation and Emotion, 36*, 208–214.

Jimeno-Ingrum, D., Berdahl, J. L., & Lucero-Wagoner, B. (2009). Stereotypes of Latinos and Whites: Do they guide evaluations in diverse work groups? *Cultural Diversity & Ethnic Minority Psychology, 15*(2), 158.

Judd, C. M., Westfall, J., & Kenny, D. A. (2012). Treating stimuli as a random factor in social psychology: A new and comprehensive solution to a pervasive but largely ignored problem. *Journal of Personality and Social Psychology, 103*(1), 54–69.

Judd, C. M., Westfall, J., & Kenny, D. A. (2017). Experiments with More Than One Random Factor: Designs, Analytic Models, and Statistical Power. *Annual Review of Psychology, 68*, 601–625.

Krippendorff, Klaus. 2019. *Content Analysis: An Introduction to Its Methodology*. 4 ed. SAGE Publications, Inc.

Laustsen, L. (2014). Decomposing the relationship between candidates' facial appearance and electoral success. *Political Behavior, 36*(4), 777–791.

Levin, J., & Thomas, A. R. (1997). Experimentally manipulating race: Perceptions of police brutality in an arrest: A research note. *Justice Quarterly, 14*(3), 577–586.

Levine, Emma E. and Maurice E. Schweitzer. 2015. "The affective and interpersonal consequences of obesity." *Organizational Behavior and Human Decision Processes* 127:66–84. https://www.sciencedirect.com/science/article/pii/S0749597815000035

Lukaszewski, A. W., Simmons, Z. L., Anderson, C., & Roney, J. R. (2016). The Role of Physical Formidability in Human Social Status Allocation. *Journal of Personality and Social Psychology, 110*(3), 385–406.

Maddan, Sean, Jeffery T. Walker and J. Mitchell Miller. 2008. "Does size really matter?: A reexamination of Sheldon's somatotypes and criminal behavior." *The Social Science Journal* 45(2):330–344. https://www.sciencedirect.com/science/article/pii/S036233190800030X

Marins, E, A Rombaldi and F Del Vecchio. 2018. "Occupational physical fitness tests for police officers–a narrative review.".

Marins, E. F., David, G. B., & Del Vecchio, F. B. (2019). Characterization of the Physical Fitness of Police Officers: A Systematic Review. *Journal of Strength and Conditioning Research, 33*(10), 2860–2874.

McCandless, B. R., Scott Persons, W., & III and Albert Roberts. (1972). Perceived opportunity, delinquency, race, and body build among delinquent youth. *Journal of Consulting and Clinical Psychology,38*(2), 281.

McNeish, D. (2018). Thanks coefficient alpha, we'll take it from here. *Psychological Methods, 23*(3), 412–433.

Melamed, D., Munn, C. W., Barry, L., Montgomery, B., & Okuwobi, O. F. (2019). Status characteristics, implicit bias, and the production of racial inequality. *American Sociological Review, 84*(6), 1013–1036.

Mize, Trenton D. 2025. "Competence Perceptions at the Intersection of Gender and Race-Ethnicity." *Social Psychology Quarterly* 88(2):271–284. https://journals.sagepub.com/home/spq

Mize, T. D., & Manago, B. (2018). The stereotype content of sexual orientation. *Social Currents, 5*(5), 458–478.

Monin, B., & Oppenheimer, D. M. (2014). The limits of direct replications and the virtues of stimulus sampling. *Social Psychology, 45*(4), 299–300.

Mueller, U., & Mazur, A. (1996). Facial dominance of West Point cadets as a predictor of later military rank. *Social forces, 74*(3), 823–850.

Nature Human Behaviour Editorial Board. (2023). Living in a brave new AI era. *Nature Human Behaviour, 7*(11), 1799.

Olivola, Christopher Y and Alexander Todorov. 2010*a*. "Elected in 100 milliseconds: Appearance-based trait inferences and voting." *Journal of nonverbal behavior* 34(2):83–110.

Olivola, Christopher Y and Alexander Todorov. 2010*b*. "Fooled by first impressions? Reexamining the diagnostic value of appearance-based inferences." *Journal of Experimental Social Psychology* 46(2):315–324.

Park, H., G. Kim, D. Lee and H.K. Kim. 2024. Can You Spot the AI-Generated Images? Distinguishing Fake Images Using Signal Detection Theory. In *Cross-Cultural Design. HCII 2024*, ed. P.L.P. Rau. Vol. 14702 of *Lecture Notes in Computer Science* Springer, Cham pp. 287–302. https://doi.org/10.1007/978-3-031-60913-8_21

Pica, E., Sheahan, C. L., Pozzulo, J., & Bennell, C. (2020). Guns, gloves, and tasers: Perceptions of police officers and their use of weapon as a function of race and gender. *Journal of Police and Criminal Psychology, 35*, 348–359.

Pingitore, Regina, Bernard L. Dugoni, R. Scott Tindale and Bonnie Spring. 1994. "Bias Against Overweight Job Applicants in a Simulated Employment Interview." *Journal of Applied Psychology* 79(6):909–917. https://www.researchgate.net/publication/15347435

Puhl, Rebecca and Kelly D Brownell. 2013. "Bias, discrimination and obesity." *Health and human rights in a changing world* pp. 581–606.

Puhl, R. M., & Brownell, K. D. (2006). Confronting and coping with weight stigma: an investigation of overweight and obese adults. *Obesity, 14*(10), 1802–1815.

Ridgeway, C. L. (2019). *Status: Why is it everywhere? Why does it matter?* Russell Sage Foundation.

Roehling, M. V. (1999). Weight-based discrimination in employment: Psychological and legal aspects. *Personnel Psychology, 52*(4), 969–1016.

Salerno, J. M., & Sanchez, J. (2020). Subjective Interpretation of "Objective" Video Evidence: Perceptions of Male Versus Female Police Officers' Use-of-Force. *Law and Human Behavior, 44*(2), 97–112.

Sheldon, William Herbert. 1949. "Varieties of delinquent youth: an introduction to constitutional psychiatry.".

Sheldon, W. H., Stevens, S. S., & Tucker, W. B. (1940). *The Varieties of Human Physique: An Introduction to Constitutional Psychology*. New York: Harper & Brothers.

Strings, S. (2019). *Fearing the Black Body: The Racial Origins of Fat Phobia*. New York, NY: New York University Press.

Todorov, A., Mandisodza, A. N., Goren, A., & Hall, C. C. (2005). Inferences of competence from faces predict election outcomes. *Science, 308*(5728), 1623–1626.

Velásquez-Salamanca, D., Martín-Pascual, M. Á., & Andreu-Sánchez, C. (2025). Interpretation of AI-Generated vs. Human-Made Images. *Journal of Imaging, 11*(7), 227. https://doi.org/10.3390/jimaging11070227

von Rueden, Christopher. 2014. The Roots and Fruits of Social Status in Small-Scale Human Societies. In *The Psychology of Social Status*, ed. Joey Cheng, Jessica L. Tracy and Cameron Anderson. New York, NY: Springer pp. 179–200.

Wagner, Michael, Matthew Harper, Ashley R. Rockwell and William Wells. 2023. "Police Chief Perceptions of Officer Physical Fitness and Barriers to Better Fitness." *International Journal of Police Science* 2(1). https://doi.org/10.56331/ijps.v2i1.7613

Weitzer, R. (2000). White, black, or blue cops? Race and citizen assessments of police officers. *Journal of Criminal Justice, 28*(4), 313–324. https://doi.org/10.1016/S0047-2352(00)00043-X

Wells, G. L. and P. D. Windschitl. 1999. "Stimulus Sampling and Social Psychological Experimentation." *Personality and Social Psychology Bulletin* 25(9):1115–1125. Original work published 1999.

Wilson, Jeremy M. and Clifford A. Grammich. 2022. "Staffing Composition in Large, U.S. Police Departments: Benchmarking Workforce Diversity." *Policing: An International Journal* 45(5):707–726.

Wu, Y., Qiu, B. R., & Swadish, S. (2026). Conducting online randomized experiments: Lessons from a policing survey using AI-Edited videos. *Journal of Experimental Criminology*. https://doi.org/10.1007/s11292-025-09722-4

Yap, A. J., Mason, M. F., & Ames, D. R. (2013). The powerful size others down: The link between power and estimates of others' size. *Journal of Experimental Social Psychology, 49*, 591–594. https://doi.org/10.1016/j.jesp.2012.10.003

Yonce, Kelsey P. 2014. Attractiveness Privilege: The Unearned Advantages of Physical Attractiveness. Master's thesis Smith College Northampton, MA: . https://scholarworks.smith.edu/theses/745

Zamudio, Cristian, John L. Grigsby and Matthew Michelsen. 2025. "RAISE: A New Method to Develop Experimental Stimuli for Advertising Research with Image Generative Artificial Intelligence." *Journal of Advertising* pp. 1–16.

Zanolie, K., van Dantzig, S., Boot, I., Wijnen, J., Schubert, T. W., Giessner, S. R., & Pecher, D. (2012). Mighty metaphors: Behavioral and ERP evidence that power shifts attention on a vertical dimension. *Brain and Cognition, 78*, 50–58. https://doi.org/10.1016/j.bandc.2011.10.006

Zeng, T. C., Cheng, J. T., & Henrich, J. (2022). Dominance in Humans. *Philosophical Transactions of the Royal Society B: Biological Sciences, 377*(1845), 20200451.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Chandler G. Robinson Chandler is a PhD Student in the Political Science Department at The University of Utah.

Ian T. Adams, PhD Dr. Adams is an Assistant Professor in the Department of Criminology & Criminal Justice at the University of South Carolina. His applied research focuses on the practical concerns of police practitioners, with a specific interest in technology, policy, behavior, and use-of-force in law enforcement.

Matthew W. Logan, PhD Dr. Logan is an Associate Professor at Texas State University School of Criminal Justice and Criminology. His primary research interests are applied and institutional corrections, and he has published in several journals across the social and behavioral sciences.

J. Pete Blair, PhD Dr. Blair is the Executive Director of the Advanced Law Enforcement Rapid Response Training (ALERRT) Center at Texas State University. His research has focused on active-shooter events, police tactics, and deception detection. His research has resulted in journal publications and books. He has been invited to speak on active shooter events and police tactics by groups across the USA and internationally.